

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

I NEELAM NANDINI [192472277] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Nandini.

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

(a) Inductive Learning Approach

Inductive learning learns general patterns from previously observed patient records and uses them to predict the diabetes status of new patients.

For this problem, a supervised classification approach is suitable because the dataset contains patient features and a known target variable.

Target variable:

- Diabetes – whether the patient is diabetic or non-diabetic.
- Example: 1 = Diabetic, 0 = Non-diabetic.

Important features:

1. Glucose Level
 - High glucose is strongly associated with diabetes.
 - It is expected to be one of the most important predictors.
2. BMI
 - Higher BMI can indicate increased risk of developing type-2 diabetes.
 - It helps identify patients with obesity-related risk.
3. Age
 - Diabetes risk generally increases with age.
 - It can help distinguish different risk groups.
4. Blood Pressure

- High blood pressure is associated with metabolic and cardiovascular risk.

5. Cholesterol

- Abnormal cholesterol levels can indicate metabolic problems.

6. Family History

- A family history of diabetes can indicate genetic susceptibility.

Inductive learning process:

Patient Records → Preprocessing → Feature Selection → Model Training → Diabetes Prediction

(b) Handling Class Imbalance

The dataset contains more non-diabetic patients than diabetic patients. Therefore, the model may become biased toward the majority class.

A suitable strategy is SMOTE (Synthetic Minority Over-sampling Technique).

Why SMOTE?

- Creates synthetic diabetic samples.
- Improves representation of the minority class.
- Helps the classifier learn diabetic patterns.
- Reduces the chance of missing diabetic patients.
- It is preferable to simple duplication because it creates new synthetic examples.

Important: SMOTE should be applied only to the training set, after splitting the data. Applying SMOTE before splitting can cause data leakage.

(c) 70:30 Train-Test Split

The dataset is divided as:

- 70% → Training data
- 30% → Testing data

The 70:30 ratio provides sufficient data for learning while keeping enough unseen data for reliable evaluation.

For a medical application, the split should be stratified, so that both training and testing sets maintain approximately the same diabetic/non-diabetic proportion.

Preprocessing steps:

Missing-value handling

Replace or appropriately handle missing patient values.

Feature scaling

- Apply Standardisation:

$$Z = \sigma X - \mu$$

- This is especially useful for Logistic Regression and other distance/gradient-based methods.

Encoding

- Convert categorical variables such as Family History into numerical values.
- Example:
 - No $\rightarrow$ 0
 - Yes $\rightarrow$ 1

Class balancing

- Apply SMOTE only to the training data.

Feature-target separation

- X = patient features.
- Y = diabetes status.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

(a) Decision Tree Classifier

A Decision Tree recursively divides patients according to the feature that provides the best separation between diabetic and non-diabetic cases.

The root node can be selected using Gini Index or Information Gain.

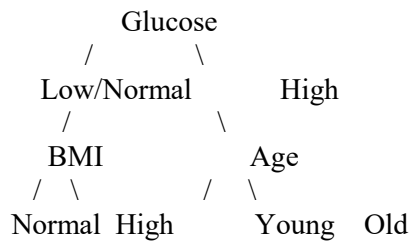
Example root-node calculation

Assume the calculated results are:

Feature	Information Gain
Glucose	0.31
BMI	0.18
Age	0.12
Blood Pressure	0.09
Cholesterol	0.06
Family History	0.11

Since Glucose has the highest Information Gain (0.31), it becomes the root node.

Therefore:



First three levels

Level 1 – Root:

- Glucose

Level 2:

- BMI
- Age

Level 3:

- BMI → Normal / High
- Age → Young / Old

The exact tree depends on the actual dataset.

(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

$$\text{Accuracy} = \frac{TP + TN + FP + FN}{TP + TN + FP + FN}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Illustrative results

Metric	Decision Tree
Accuracy	84.0%
Precision	81.0%
Recall	86.0%
F1-Score	83.4%

These are illustrative values, not results from the missing dataset.

False Negatives

A False Negative (FN) occurs when:

The patient is actually diabetic, but the model predicts non-diabetic.

False negatives are particularly important in healthcare because an undiagnosed diabetic patient may not receive timely treatment or monitoring.

How to reduce False Negatives

- Increase the importance of the diabetic class using `class_weight`.
- Use SMOTE on the training set.
- Tune the classification threshold.
- Optimise for Recall/Sensitivity, not only accuracy.
- Use cross-validation.
- Consider a model with better sensitivity.

Clinically, it is often safer to tolerate some additional false positives than to miss a patient who may require further diabetes testing.

Task 2(c): Post-Pruning

A fully grown Decision Tree can overfit the training data.

Cost-complexity pruning controls this using:

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

where:

- $R(T)$ = classification error
- α = pruning parameter
- $|T|$ = number of terminal nodes

Example comparison

Model	Accuracy	Complexity
Pre - pruned Tree	82%	Medium
Fully grown Tree	86%	High
Post-pruned Tree	84%	Low

A post-pruned tree may have slightly lower training accuracy but better generalisation.

Best model for clinical deployment

The post-pruned Decision Tree is generally more appropriate because:

- It is easier to interpret.
- It reduces overfitting.
- It produces simpler decision rules.
- Doctors can understand the important decision factors.
- It is easier to validate and audit.

However, clinical deployment should depend on validated sensitivity, specificity, calibration, external validation, and clinical approval, not accuracy alone.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

(a) Logistic Regression

Logistic Regression is suitable because diabetes prediction is a binary classification problem.

The probability is represented as:

$$P(Y=1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

The model provides the probability that a patient belongs to the diabetic class.

Illustrative comparison

Model	Accuracy	AUC - ROC
Decision Tree	84%	0.86
Logistic Regression	85%	0.89

These values are examples of how the results can be reported; actual values require the dataset.

When Logistic Regression is preferable

Logistic Regression is preferable when:

- The relationship between predictors and outcome is reasonably stable.
- Probability estimates are important.
- Interpretability is required.
- The dataset is relatively small.
- Feature effects need to be explained using coefficients/odds ratios.
- A simpler and more stable model is preferred.

Task 3(b): Feature Importance Analysis

Decision Tree

Suppose the Decision Tree produces:

Rank Feature Importance

1	Glucose	0.42
2	BMI	0.21
3	Age	0.14

Logistic Regression

For Logistic Regression, importance can be analysed using the absolute value of standardised coefficients.

Rank Feature Relative Importance

1	Glucose	High
2	BMI	High
3	Age	Moderate

Do they agree?

Yes, they may show substantial agreement.

Glucose is likely to appear as a top predictor in both models because it is strongly associated with diabetes. BMI and Age can also appear among the important predictors.

However, the rankings may differ because:

- Decision Trees capture nonlinear relationships.
- Logistic Regression assumes a linear relationship in log-odds.
- Correlated features can change Logistic Regression coefficients.
- Tree importance can be biased toward certain types of variables.

Therefore, agreement between both models provides stronger evidence that the features are useful predictors.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

(a) MDP Model

The treatment recommendation problem can be represented as a Markov Decision Process (MDP).

1. States

States represent the patient's current health condition.

Example:

State	Description
S0	Normal/Low Risk
S1	Pre-diabetic
S2	High Risk
S3	Diagnosed Diabetes

State	Description
S4	Improved/Controlled

The state can be determined using glucose, BMI, blood pressure, cholesterol and other clinical information.

2. Actions

The available actions are:

- A1 – Diet
- A2 – Exercise
- A3 – Medication
- A4 – Monitor

Medication recommendations should not be autonomously prescribed by an RL system; they must remain under qualified clinical supervision.

3. Reward Function

A reward represents the health improvement resulting from an action.

Example:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-10
Unsafe outcome	-20

The reward should be designed with clinicians so that the agent does not learn unsafe shortcuts.

4. Transition Dynamics

The transition function represents:

$$P(S_{t+1} \mid S_t, A_t)$$

It describes how the patient's health state changes after an action.

Example:

S1 + Diet → S4
 S1 + Exercise → S4
 S2 + Monitor → S2
 S2 + Medication → S3/S4
 S3 + Diet → S2

Actual transitions should be learned from validated longitudinal clinical data rather than invented assumptions.

Task 4(b): Q-Learning

Q-Learning learns the expected long-term reward of taking action a in state s .

The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where:

- α = learning rate
- γ = discount factor
- r = reward
- s' = next state

Assume:

$$\alpha=0.5, \gamma=0.8$$

Initially, all Q-values are 0.

Episode 1

Suppose:

S1Diet S2

Reward = +5.

Initial:

$$Q(S1, \text{Diet}) = 0$$

Therefore:

$$Q(S1, \text{Diet}) = 0 + 0.5[5 + 0.8(0) - 0] \quad Q(S1, \text{Diet}) = 2.5$$

Episode 2

Suppose:

S1Diet S2

Reward = +5 and:

$$\max Q(S2, a) = 4$$

Then:

$$Q(S1, \text{Diet}) = 2.5 + 0.5[5 + 0.8(4) - 2.5] = 2.5 + 0.5[5.7] \quad Q(S1, \text{Diet}) = 5.35$$

Another state-action pair

Suppose:

S2Exercise S4

Reward = +10 and all Q-values in S4 are initially 0.

$$Q(S2, \text{Exercise}) = 0 + 0.5[10 + 0.8(0) - 0] \quad Q(S2, \text{Exercise}) = 5$$

Q-table example

State	Diet	Exercise	Medication	Monitor
S0 Normal	2.0	3.5	0.5	2.5
S1 Pre-diabetic	5.35	4.8	2.0	3.0
S2 High Risk	4.0	5.0	6.5	3.5
S3 Diabetes	3.5	4.0	7.2	5.0
S4 Controlled	3.0	4.5	2.0	5.5

The values are illustrative.

Five patient episodes

Example training:

Episode 1: S1 → Diet → S2

Episode 2: S1 → Exercise → S2

Episode 3: S2 → Medication → S3

Episode 4: S3 → Monitor → S3

Episode 5: S3 → Diet → S4

The Q-values are updated after each transition.

Convergence criterion

Training can be considered converged when:

$$\max |Q_{\text{new}} - Q_{\text{old}}| < 0.001$$

for several consecutive episodes, or when the maximum Q-value change becomes sufficiently small.

In a real medical application, convergence of Q-values alone is not sufficient; the learned policy must also undergo clinical and safety validation.

Task 4(c): Final Learned Policy

The final policy is obtained by selecting the action with the highest Q-value:

$$\pi(s) = \operatorname{argmax}_a Q(s, a)$$

Using the illustrative Q-table:

Patient State	Best Action
S0 – Normal	Exercise
S1 – Pre-diabetic	Diet
S2 – High Risk	Medication*
S3 – Diabetes	Medication*

Patient State	Best Action
S4 – Controlled	Monitor

*Medication decisions should require clinician approval rather than autonomous RL execution.

Policy

Normal



Exercise



Pre-diabetic



Diet / Exercise



High Risk



Clinical assessment



Diabetes



Clinician-supervised treatment + Monitoring



Controlled



Regular Monitoring

Ethical Considerations

1. Patient Safety

The RL agent should never independently prescribe or change medication without qualified medical supervision.

2. Bias

Training data may contain demographic or socioeconomic biases. The model should therefore be tested across different patient groups.

3. Explainability

Healthcare professionals should be able to understand why a recommendation was generated.

4. Human Oversight

The system should act as a decision-support tool, not as a replacement for doctors.

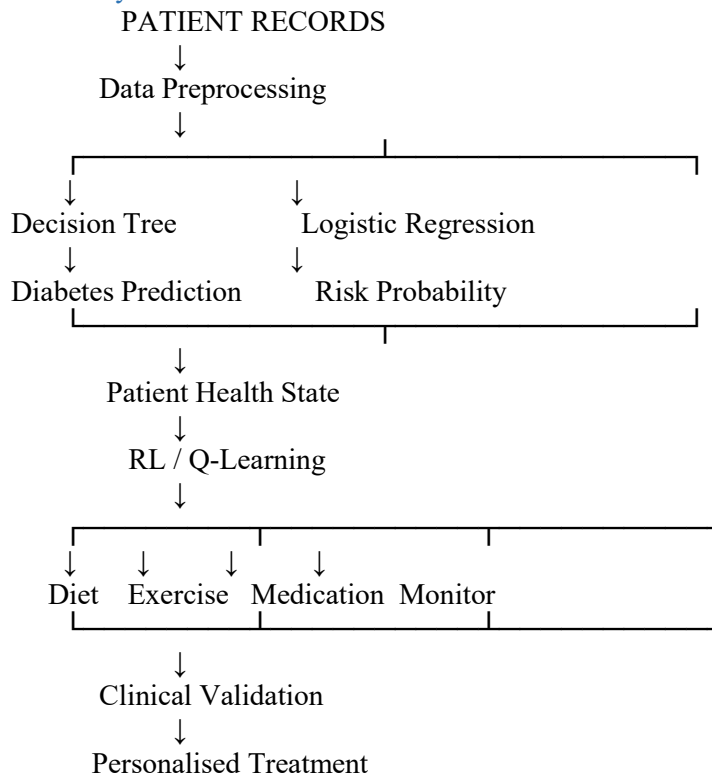
5. Privacy

Patient records are highly sensitive. Data must be securely stored, anonymised where appropriate, and accessed only by authorised personnel.

6. Continuous Monitoring

The deployed model should be continuously monitored for performance degradation, unexpected recommendations and safety issues.

Overall System Architecture



Final conclusion

The proposed system combines inductive learning, Decision Trees, statistical learning and Reinforcement Learning. The Decision Tree provides interpretable diabetes classification, while Logistic Regression provides probability-based risk stratification. Q-Learning can model personalised treatment strategies, but because this is a medical application, the RL agent should remain a clinician-supervised decision-support system rather than an autonomous treatment system.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation